

MKTG 891 Top/Marketing (Independent Study)

Corn/Maize leaf disease prediction

Steps Followed for Model Development:

1. Data Preprocessing

- **What was done:**
 - **Image Resizing:** The input images from the dataset were likely resized to match the input size required by the ResNet18 and MobileNetV2 models (e.g., 224x224 pixels).
 - **Normalization:** Pixel values were scaled between 0 and 1 or standardized to have zero mean and unit variance, which helps the model converge faster during training.
 - **Data Augmentation:** Applied techniques like flipping, rotation, blurring, increase the size of the dataset and introduce variability.
- **Why?** These preprocessing steps ensure that the images fed into the model are of consistent size and distribution, which improves the model's ability to generalize and learn effectively. Augmentation helps prevent overfitting by making the model robust to variations in the data.

2. Building the Model

- **ResNet18:**
 - This model uses **residual blocks**, which allow the network to skip some layers by introducing shortcut connections. This helps address the **vanishing gradient problem**, where deeper networks struggle to propagate information backward during training. As a result, ResNet18 can have more layers while still being easy to train.
- **MobileNetV2:**
 - This model is designed for mobile and resource-constrained environments. It uses **depthwise separable convolutions**, reducing the number of parameters and computations needed. This makes MobileNetV2 faster and more efficient, but it sacrifices a bit of accuracy compared to larger models like ResNet18.
- **DenseNet169:**
 - This model uses dense connections, where each layer has direct connections to subsequent layers, promoting **feature reuse** and reducing the number of parameters. This dense connectivity helps alleviate the vanishing gradient issue and achieves better flow of information and gradients throughout the network.

- **VGG16:**
 - VGG16 is a deep network with 16 layers, characterized by its simplicity of using **3x3 convolution layers** stacked on top of each other, which enhances its feature extraction capabilities.
- **Why?** These architectures were chosen for their strong performance in image classification tasks. ResNet18 provides high accuracy through its deep residual structure, ideal for complex feature extraction. MobileNetV2, designed with depthwise separable convolutions, prioritizes computational efficiency and speed, making it suitable for resource-constrained environments. DenseNet169 leverages dense connections for improved gradient flow and feature reuse, enabling it to achieve high accuracy with fewer parameters. VGG16, while computationally intensive, is a robust model with a deep, layered architecture that excels in high-accuracy classification. Using transfer learning across these models, which start with pretrained weights from a large dataset like ImageNet, provides a strong foundational understanding of general image features, allowing each model to adapt effectively to the specific task of disease classification with less data and reduced training time.

3. Fine-Tuning the Model

- **What was done:**
 - **Freezing Early Layers:** The initial layers of both **ResNet18** and **MobileNetV2** were frozen to prevent their weights from being updated during training. These layers are already well-suited to general image recognition tasks from the ImageNet pre-training.
 - **Custom Classification Layer:** We replaced the original fully connected (dense) layer with a new one that matches the number of classes in your corn disease dataset. For example, if you have three types of corn diseases, the final layer would output three classes.
- **Why?** Freezing these layers helps retain the learned features from ImageNet and allows faster training by focusing on updating only the task-specific layers. By only training the later layers, you prevent overfitting and reduce the training time, as the model doesn't need to learn basic image features from scratch.

4. Training the Model

- **What was done:**
 - **Defining a Loss Function:** used categorical cross-entropy loss, which is well-suited for multi-class classification problems like this.

- **Optimizer:** used an optimizer like Adam, which adjusts the learning rate adaptively during training, helping the model converge faster.
- **Why?** Training is essential for the model to learn the patterns associated with different corn diseases. By adjusting the weights in the network, the model minimizes the difference between its predictions and the true labels.

5. Evaluation of the Model

- **What you did:**
 - **Accuracy:** The percentage of correct predictions across the test data.
 - **Precision:** The proportion of true positive predictions out of all positive predictions (i.e., when the model predicts a disease, how often it's correct).
 - **Recall (Sensitivity):** The proportion of actual positives correctly identified by the model (i.e., how well the model identifies the disease when it's present).
 - **F1 Score:** The harmonic mean of precision and recall, balancing the two in cases where you want a trade-off.
 - **Confusion Matrix:** used this to visualize how well the model is classifying each disease category, showing the number of true positives, false positives, true negatives, and false negatives.
- **Why?** These metrics give a complete picture of how well the model performs, not just in terms of accuracy but also in its ability to handle false positives and false negatives, which are crucial in disease prediction scenarios.

6. Model Performance Comparison

Model	Accuracy	Precision	Recall (Sensitivity)	F1 Score	Specificity	Weighted Avg Precision	Weighted Avg Recall	Weighted Avg F1 Score
DenseNet169	98.34%	0.98	0.98	0.98	0.98	0.98	0.98	0.98
MobileNetV2	96.44%	0.96	0.95	0.95	0.95	0.96	0.96	0.96
VGG16	96.21%	0.95	0.95	0.95	0.95	0.96	0.96	0.96
ResNet18	89.10%	0.90	0.84	0.85	0.84	0.91	0.89	0.88

Analysis

- **DenseNet169:** This model achieved the highest accuracy (98.34%) among all models, with precision, recall, and F1 scores around 0.98, indicating excellent performance across all metrics. Its architecture enables efficient feature reuse and gradient flow, contributing to its high accuracy and balanced performance in disease detection.
- **MobileNetV2:** With an accuracy of 96.44%, MobileNetV2 combines strong performance with computational efficiency, making it a good choice for applications where resources are limited. The model's slightly lower precision and recall compared to DenseNet169 still yield a robust F1 score of 0.95, providing reliable results while maintaining speed.
- **ResNet18:** ResNet18 has a lower accuracy of 89.10% and lower recall and F1 scores, reflecting some limitations in handling this dataset compared to the other models. While ResNet's residual architecture is effective for general feature learning, it may require further tuning or may be less suited for this particular classification task due to its comparatively lower sensitivity and specificity.
- **VGG16:** Achieving a close second in accuracy at 96.21%, VGG16 shows high precision, recall, and F1 scores, similar to MobileNetV2. Although it is computationally intensive, its high accuracy and robust classification performance make it suitable for use cases where computational resources are not as constrained.

Conclusion: **DenseNet169** stands out as the most accurate model with balanced performance across all metrics, followed closely by **MobileNetV2**, which offers a strong balance between efficiency and accuracy. **VGG16** also provides high accuracy but is more computationally intensive, making it suitable when resources allow. **ResNet18** may require further tuning or could be best suited for scenarios where simplicity or speed is prioritized over maximum accuracy.